

A Technical Introduction to Expectation maximization algorithm

Section 1

Maximization step: Choose θ to maximize $F(\theta)$: $\theta(t+1) = \arg \max_{\theta} F(\theta(t), \theta)$

Section 2

$$Q(\theta | \theta(t)) = E Z | X = x; \theta(t) \left[\log L(\theta; x, Z) \right] \\ = E Z | X = x; \theta(t) \left[\log \prod_{i=1}^n L(\theta; x_i, Z_i) \right] = E Z | X = x; \theta(t) \left[\sum_{i=1}^n \log L(\theta; x_i, Z_i) \right] = \sum_{i=1}^n E Z_i | X_i = x_i; \theta(t) \left[\log L(\theta; x_i, Z_i) \right] = \sum_{i=1}^n \sum_{j=1}^2 P(Z_i = j | X_i = x_i; \theta(t)) \log L(\theta; x_i, j) \\ = \sum_{i=1}^n \sum_{j=1}^2 T_{j,i}(t) \left[\log \tau_j - \frac{1}{2} \log \left| \Sigma_j \right| - \frac{1}{2} (x_i - \mu_j)^T \Sigma_j^{-1} (x_i - \mu_j) - \frac{d}{2} \log (2\pi) \right].$$

$$\begin{aligned} Q(\theta | \theta(t)) &= \operatorname{E} \left[\log L(\theta; \mathbf{X}, \mathbf{Z}) \mid \theta(t) \right] \\ &= \operatorname{E} \left[\log \prod_{i=1}^n L(\theta; \mathbf{x}_i, \mathbf{z}_i) \mid \theta(t) \right] \\ &= \sum_{i=1}^n \operatorname{E} \left[\log L(\theta; \mathbf{x}_i, \mathbf{z}_i) \mid \theta(t) \right] \\ &= \sum_{i=1}^n \sum_{j=1}^2 P(Z_i = j | \mathbf{x}_i, \theta(t)) \log L(\theta; \mathbf{x}_i, j) \\ &= \sum_{i=1}^n \sum_{j=1}^2 T_{j,i}(t) \left[\log \tau_j - \frac{1}{2} \log |\Sigma_j| - \frac{1}{2} (\mathbf{x}_i - \mu_j)^T \Sigma_j^{-1} (\mathbf{x}_i - \mu_j) - \frac{d}{2} \log (2\pi) \right] \end{aligned}$$

Section 3

E-step Operate a Kalman filter or a minimum-variance smoother designed with current parameter estimates to obtain updated state estimates. M-step Use the filtered or smoothed state estimates within maximum-likelihood calculations to obtain updated parameter estimates. Suppose that a Kalman filter or minimum-variance smoother operates on measurements of a single-input-single-output system that possess additive white noise. An updated measurement noise variance estimate can be obtained from the maximum likelihood calculation

Section 4

In words, choosing θ to improve $Q(\theta | \theta(t))$ causes $\log p(\theta | \theta(t))$ to improve at least as much.

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Section 5

$$\tau_j(t+1) = \sum_{i=1}^n T_{j,i}(t) \left[\log \tau_j - \frac{1}{2} \log |\Sigma_j| - \frac{1}{2} (\mathbf{x}_i - \mu_j)^T \Sigma_j^{-1} (\mathbf{x}_i - \mu_j) - \frac{d}{2} \log (2\pi) \right]$$

Section 6

where $\hat{x}_k$ are scalar output estimates calculated by a filter or a smoother from N scalar measurements z_k . The above update can also be applied to updating a Poisson measurement noise intensity. Similarly, for a first-order auto-regressive process, an updated process noise variance estimate can be calculated by

Section 7

$$(\mu_1(t+1), \Sigma_1(t+1)) = \arg \max_{\mu_1, \Sigma_1} Q(\theta | \theta(t)) = \arg \max_{\mu_1, \Sigma_1} \sum_{i=1}^n T_{1,i}(t) \left[-\frac{1}{2} \log |\Sigma_1| - \frac{1}{2} (\mathbf{x}_i - \mu_1)^T \Sigma_1^{-1} (\mathbf{x}_i - \mu_1) \right]$$

Section 8

Proof of correctness Expectation-Maximization works to improve $Q(\theta | \theta(t))$ rather than directly improving $\log p(\theta | \theta(t))$. Here it is shown that improvements to the former imply improvements to the latter. For any Z with non-zero probability $p(Z | X, \theta)$, we can write

Section 9

The symbols Given the statistical model which generates a set X of observed data, a set of unobserved latent data or missing values Z , and a vector of unknown parameters θ , along with a likelihood function $L(\theta; X, Z) = p(X, Z | \theta)$, the maximum likelihood estimate (MLE) of the

unknown parameters is determined by maximizing the marginal likelihood of the observed data

Section 10

where $\mathbb{I}$ is an indicator function and f is the probability density function of a multivariate normal. In the last equality, for each i , one indicator $\mathbb{I}(z_i = j)$ is equal to zero, and one indicator is equal to one. The inner sum thus reduces to one term.